Yaghoub Rahimi

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Academic Appointments

- Postdoctoral Researcher in Mathematics — August 2025–Present
  Louisiana State University, Department of Mathematics
  - Advisor: Rui Han.

Education

- Ph.D. in Mathematics — August 2019 - July 2025
  Georgia Institute of Technology
  - Supervised by Dr. Michael Lacey
- M.Sc. in Mathematics — August 2016 - July 2019
  The University of Texas at Dallas
  - Supervised by Dr. Yifei Lou
- M.Sc. in Mathematics — September 2013 - January 2016
  Sharif University of Technology
  - Supervised by Dr. A. Daneshgar
- B.Sc. in Applied Mathematics — September 2009 - 2013
  Shiraz University

Research Interests

- Analytic Number Theory and Additive Problems
- Discrete Harmonic Analysis and Fourier-Analytic Methods
- Restricted-Digit and Cantor-Type Arithmetic Sets
- Additive Combinatorics
- Sparse Signal Recovery and Nonconvex Optimization (earlier work)

Publications and Preprints

Preprints

1. *Density Bounds in Hua's Theorem for Prime Squares*
   with Michael T. Lacey, Hamed Mousavi, and Manasa N. Vempati.
   Preprint, 2026. arXiv:2610.22747.
   arXiv | PDF | Formalization
2. *Primes with Restricted-Digit Differences*
   with Rui Han and Fan Yang.
   Preprint, 2026. arXiv:2609.05810.
   arXiv | PDF
3. *On the small denominator problem for generalized Minkowski–Funk transforms*
   with Rui Han.
   Preprint, 2026. arXiv:2601.09547.
   arXiv | PDF

4. *Averages over the Gaussian Primes: Goldbach's Conjecture and Improving Estimates*
with Christina Giannitsi, Ben Krause, Michael T. Lacey, and Hamed Mousavi.
Preprint, 2023. arXiv:2309.14249.
arXiv | PDF

Published papers

1. *A density theorem for higher-order sums of prime numbers*
with Michael T. Lacey, Hamed Mousavi, and Manasa N. Vempati.
Research in the Mathematical Sciences 13, Article 17, 2026.
arXiv | Journal | PDF

2. *A lifted ℓ_1 framework for sparse recovery*
with Sung Ha Kang and Yifei Lou.
Information and Inference: A Journal of the IMA 13(1), iaad055, 2024.
arXiv | Journal | PDF

3. *Endpoint ℓ^r improving estimates for prime averages*
with Michael T. Lacey and Hamed Mousavi.
Mathematical Research Letters 29(6), 1767–1791, 2022.
arXiv | Journal | PDF

4. *Improving and maximal inequalities for primes in progressions*
with Christina Giannitsi, Michael T. Lacey, and Hamed Mousavi.
Banach Journal of Mathematical Analysis 16(3), Article 42, 2022.
arXiv | Journal | PDF

5. *Accelerated schemes for the L_1/L_2 minimization*
with Chao Wang, Ming Yan, and Yifei Lou.
IEEE Transactions on Signal Processing 68, 2660–2669, 2020.
arXiv | Journal | PDF | Code

6. *A scale-invariant approach for sparse signal recovery*
with Chao Wang, Hongbo Dong, and Yifei Lou.
SIAM Journal on Scientific Computing 41(6), A3649–A3672, 2019.
arXiv | Journal | PDF | Code

Selected Talks

- **A Density Theorem for Higher-Order Sums of Prime Numbers**, Talk at the Harmonic Analysis Seminar, Louisiana State University, Baton Rouge, LA, May 6, 2026.
- **A Density Theorem for Higher-Order Sums of Prime Numbers**, Talk at the Southern Regional Harmonic Analysis Conference, Louisiana State University, Baton Rouge, LA, November 1, 2025.
- **Improving Averages over the Prime Numbers and Goldbach's Conjecture**, Ph.D. dissertation defense, Georgia Institute of Technology, July 3, 2025.
- **A Density Theorem for Higher-Order Sums of Prime Numbers**, Talk at Southern Regional Number Theory Conference 2025, Louisiana State University, Baton Rouge, LA, March 29, 2025.
- **Averages over the Gaussian Primes: Improving Estimates and Goldbach's Conjecture**, Talk at the Harmonic Analysis Seminar, Louisiana State University (online), February 28, 2024.
- **Explicit Construction of RIP Matrices and Related Problems**, Talk at Heat Flow, two-point inequalities, and beyond Summer School, Oct 2023, Online Workshop.
- **Endpoint ℓ^p Improving Estimates for Prime Averages**, Talk at Combinatorial and Additive Number Theory Seminar (CANT), May 28, 2021, Online Conference.
- **A Scale-Invariant Approach for Sparse Signal Recovery**, Poster Presentation, The University of Texas at Dallas, USA, Spring 2019
- **Training Neural Networks**, Talk in the Image and Signal Analysis Journal Club, The University of Texas at Dallas, USA, Spring 2019

- **Sparse Signal Recovery Using** ℓ_1/ℓ_2, Talk in the Image and Signal Analysis Journal Club, The University of Texas at Dallas, USA, Fall 2018
- **On Logistic Regression**, Talk in the Image and Signal Analysis Journal Club, The University of Texas at Dallas, USA, Fall 2017
- **Fast Single Image Super-Resolution Using Sparse Gaussian Process Regression**, Talk in the Image and Signal Analysis Journal Club, The University of Texas at Dallas, USA, Spring 2017
- **Hadamard Matrices and Latin Square**, Talk at the Sharif University of Technology, Iran , Spring 2015
- **An Introduction to Deep Learning**, Talk at the Sharif University of Technology, Iran, Fall 2014

Professional Service

- **Organizer**, *Recent Advances in Discrete Harmonic Analysis*, special session at the AMS Spring 2027 Southeastern Sectional Meeting, University of Alabama at Birmingham, March 20–21, 2027 *(upcoming)*.
  Organized with Fan Yang and Hongki Jung (all Louisiana State University).

Honors

- Early Career Mathematician Travel Grant (up to $2,400), ICM 2026, administered by the American Mathematical Society and funded by the National Science Foundation, 2026.
- Outstanding TA Award, Georgia Institute of Technology, 2025.
- Research Assistant supported by NSF Grant (M. Lacey, DMS-1949206), Georgia Institute of Technology, USA, Summer 2021.
- Research prize supported by NSF Grant (M. Lacey), $2500, Georgia Institute of Technology, Spring 2021
- David L. Brown Fellowship Award, $500, Georgia Institute of Technology, USA, April 2020
- Research Assistant supported by NSF Grant (Yifei Lou. DMS-1522786), The University of Texas at Dallas, 2017-2018.
- Third Prize (Bronze Medal) in 20th International Mathematics Competition for University Students, American University in Bulgaria, Blagoevgrad, Bulgaria, August 2013
- Gold Medal in 37th Iran Mathematics Competition, Semnan University, Semnan, Iran, February 2013
- 5th Place in 18th Iran Mathematics Olympiad for University Student, Tarbiat Modares University, Tehran, Iran, July 2013
- 13th Place (27th team) in 37th ACM International Collegiate Programming Contest (ACM-ICPC) Asia Region - Tehran Site, Sharif University of Technology, Tehran, Iran, December 2012
- 2nd Place in Shiraz ACM-ICPC Contest, Shiraz University, Shiraz, Iran, October 2012

Experience & Activities

- Research Assistant, Supervisor: Dr. Michael Lacey, Georgia Institute of Technology, 2023–2024
- Teaching Assistant, Georgia Institute of Technology, Fall 2019–Spring 2025
- Research Assistant, Supervisor: Dr. Yifei Lou, The University of Texas at Dallas, 2017–2018
- Teaching Assistant, The University of Texas at Dallas, 2016–2017, 2018–2019
- Student Worker, The University of Texas at Dallas, Summer 2019, Summer 2018
- Active Mentor in the Directed Research Program (DRP), the University of Texas at Dallas, Fall 2018, Spring 2019, Fall 2019.
- Collaboration with "Kolbeh Afarinesh Fekr" Company, Games and Mathematics Exhibitions, Tehran, Iran, 2015
- Instructor of Mathematics Olympiad in High Schools, Shiraz, Iran, 2012
- Coach of the Shiraz University Team in 39th Iranian Mathematics Competition, Yazd University, Yazd, Iran, May 2015

- Collaboration with the Mathematics Society at Shiraz University, Shiraz, Iran, 2012
- Teaching Assistant, Shiraz University, 2011–2012, 2012–2013
- Design and Implementation of "Storage System of a Market" (Using Python)
- Implementation of Snake Game (Pygame and OpenGL C++)

Teaching Experience

- MATH 2057, Multidimensional Calculus, Fall 2025, Spring 2026, and Fall 2026, Louisiana State University
- MATH 1554, Linear Algebra, Fall 2023, Georgia Institute of Technology
- MATH 2551, Multivariable Calculus, Summer 2023, Georgia Institute of Technology

Teaching Assistant Experience

- Georgia Institute of Technology (Fall 2019–Spring 2025):
  - Differential Equations (MATH 2552, Fall 2019), Algebra I and II, Multivariable Calculus.
- The University of Texas at Dallas (Fall 2016–Summer 2019):
  - Topology, Differential Calculus, Applied Calculus I and II, Calculus I and II.
- Shiraz University (Fall 2010–Spring 2013):
  - Foundation of Geometry, Linear Algebra I, Computer Foundation and Programming, Elementary Matrix and Linear Algebra, Foundation of Mathematics.

Computer Skills

- C/C++, Python, MATLAB, R, Mathematica

Membership

- Member of the American Mathematical Society, 2019–present
- Member of the Organizing Committee, Workshop on Recent Developments on Mathematical/ Statistical Approaches in Data Science (MSDAS), Department of Mathematical Sciences, The University of Texas at Dallas, May 31–June 2, 2019.
- Member of Image and Signal Analysis Journal Club, UTD, 2016-2019